

Laboratory Guide – Flame Test

Introduction

When heated in a flame, some chemical elements emit light of a characteristic color. The energy supplied by the flame is released in the form of photons with a specific wavelength, corresponding to light of a precise color.

According to the **Bohr atomic model** (and more generally all atomic models in which electrons are arranged in “quantized” energy levels), atoms of a metal exposed to a flame excite some of their electrons to higher energy levels. When these electrons return to the ground state, they emit excess energy as photons with characteristic wavelengths, different for each metal. This produces distinct flame colors that allow the identification of the tested metal.

Objectives

To demonstrate the behavior of different metals in the flame test and identify them based on the characteristic color each element gives to the flame.

Materials and Equipment

- Alcohol burner or Bunsen burner
 - Platinum wire loop (or a homemade loop made from nickel-chromium wire)
 - Test tube holder (to hold the loop)
 - A beaker (for hydrochloric acid)
 - Hydrochloric acid
 - Salts of different elements: sodium chloride, copper sulfate, potassium chloride (or another potassium salt), calcium chloride, etc.
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Flame Test Procedure

1. Before each test, clean the platinum (or nickel-chromium) loop by dipping it in hydrochloric acid and then placing it in the flame to remove any residues that could affect the results.
2. Take a small amount of a sodium salt (sodium chloride, table salt) with the loop and bring it into the flame: after a few seconds, the flame turns bright yellow.
3. Clean the loop again with hydrochloric acid and flame.

4. Repeat the test with a potassium salt: the flame turns violet (more visible when observed through cobalt blue glass).
 5. With copper sulfate, the flame becomes green.
 6. With a calcium salt, the flame becomes brick red; lithium produces crimson red; barium gives yellow-green; strontium gives intense red.
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Conclusions

With the flame test, it is possible to identify different metal cations based on the characteristic color they impart to the flame.

Metals that do not visibly color the flame still emit photons, but with wavelengths in the infrared range, which are not visible to the naked eye.

For this experiment (which, like all experiments involving an open flame, must be carried out under strict teacher supervision), metal chlorides are particularly suitable, as they are more volatile.